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The Impact of Bank Financing on Municipalities' Bond Issuance and the Real Economy

Contents

1	Big picture	3
2	Data and measurement (institutional context and key objects)	4
2.1	Institutional background: why bank qualification matters	4
2.2	Tax treatment for banks	5
2.3	Policy shock under ARRA	5
2.4	Data sources	5
2.5	Main samples and time windows	6
2.6	How treated and control municipalities are defined	6
2.7	Main county-level outcome variables	6
2.8	Bank exposure measure for the lending analysis	7
2.9	Descriptive facts	7
3	Models and empirical specifications (all equations + interpretation)	8
3.1	Conceptual framework: a tax-induced notch in municipal financing	8
3.2	Bunching estimation around the \$10 million cutoff	8
3.3	County-level first stage and second stage	9
3.4	Event-study / dynamic specification	9
3.5	Bank municipal-bond holdings regression	10
3.6	Private-lending regression	10
3.7	How the job multiplier is computed	10
4	Identification and empirical design (what makes the estimates credible)	11
4.1	Why the qualification threshold is a useful source of variation	11
4.2	Why treated and control counties are plausible counterfactuals	11
4.3	Why the results are not just the housing bust or broader recession severity	11
4.4	Why the no-effect for larger newly eligible issuers matters	12
4.5	Why the employment IV should be read as a local multiplier	12
4.6	Why the bank crowding-out design is credible	12
4.7	What the paper cannot fully disentangle	13
5	Tables and figures	13

5.1	Fig. 1 and Fig. 3: market segmentation and borrowing costs	13
5.2	Figs. 2, 4, and 5 plus Table 1: bunching around \$10 million	14
5.3	Table 2 and Fig. 6: borrowing dynamics	15
5.4	Table 3, Fig. 7, and Table 4: treated versus control counties	16
5.5	Table 5 and Fig. 8: first-stage issuance response	17
5.6	Tables 6–8 and Figs. 9–10: employment effects and robustness	17
5.7	Tables 9–10 and Fig. 11: bank holdings and crowding out	19
5.8	Section 7 aggregate magnitudes: how large was the policy overall?	20
6	Short summary	20
7	Conclusion	20
7.1	Core conclusion (what the paper establishes)	20
7.2	Economic intuition	21
7.3	Takeaway	21

1 Big picture

- **Question.** When the federal government makes municipal bonds more tax-advantaged for banks, do small local governments borrow more during a recession? Does that extra borrowing translate into employment gains, and does it crowd out private lending by banks?
- **Setting.** Before 2009, a municipality could designate its debt as *bank-qualified* only if total tax-exempt issuance in the calendar year stayed at or below **\$10 million**. The American Recovery and Reinvestment Act (ARRA) temporarily raised that threshold to **\$30 million** for **2009–2010**.
- **Core empirical challenge.** Counties that borrow more after 2009 may also be counties that were hit harder by the Great Recession or the housing bust. Likewise, banks may change mortgage or small-business lending because of local loan demand rather than because they reallocated balance-sheet capacity toward municipal bonds.
- **Main conceptual contribution.** The paper shows that a tax discontinuity on the investor side creates a segmented municipal bond market. That segmentation changes municipal borrowing costs, distorts issuance around the qualification cutoff, and makes investor-side tax incentives economically powerful for local real activity.
- **Mechanism.** Banks strongly prefer bank-qualified bonds because those bonds preserve most of the interest-expense deduction that would otherwise be lost when banks hold tax-exempt municipal debt. Municipalities near the threshold therefore have an incentive to keep issuance under the cap to retain access to bank financing.
- **Main empirical design.** The paper combines four pieces of evidence:
 - aggregate evidence on banks’ holdings of qualified versus non-qualified bonds and on spreads around the cutoff;
 - a *bunching* design around the pre-2009 \$10 million notch;
 - a county-level difference-in-differences / 2SLS design using the pre-2009 share of bunching issuers in a county interacted with the 2009–2010 policy window;
 - a bank-level Bartik-like exposure design to study municipal bond holdings and crowding out of private lending.
- **Main market-segmentation result.** Banks hold mostly bank-qualified bonds, not non-qualified ones: qualified holdings are roughly **4–5% of bank assets**, whereas non-qualified bonds are only a bit above **0.5%** of assets.
- **Main bunching result.** Prior to 2009, municipalities bunch sharply just below the \$10 million bank-qualification cutoff. The preferred bunching estimate implies that the average marginal bunching issuer reduces annual debt issuance by about **2.93%** to stay qualified.
- **Main borrowing result.** In the preferred first stage, a one-standard-deviation increase in county exposure to previously constrained issuers raises bank-qualified municipal issuance by roughly **\$2.7 million** during 2009–2010.

- **Main real effect.** Each additional **\$1 million** of instrumented bank-qualified borrowing creates or saves about **22 jobs per year**.
- **Sectoral result.** The employment effect is concentrated in **private-sector** jobs, especially **services**; there is little evidence of a government-employment response and no strong evidence for goods-producing sectors.
- **Key falsification / scope result.** Municipal issuers in the **\$10–\$30 million** range that became newly eligible in 2009 do *not* increase borrowing. The policy matters mainly for municipalities that were previously constrained by the \$10 million cap.
- **Crowding-out result.** Banks more exposed to local bank-qualified issuance increase municipal-bond holdings, and there is moderate evidence that they reduce **mortgage origination**. The evidence for crowding out of small-business lending is weak.
- **Aggregate takeaway.** The paper estimates that the policy induced about **\$3.16 billion** in extra bank-qualified debt nationwide, implying roughly **70,000 jobs** created or saved, at an implied local-government cost of about **\$45,500 per job-year**.
- **Economic takeaway.** A tax-based credit policy aimed at investors, rather than direct grants to governments, can still have meaningful real effects in a recession. But it works through a segmented financing channel and comes with some crowding out on bank balance sheets.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background: why bank qualification matters

- Municipal bonds are widely used to finance public services and infrastructure, and more than **93%** of municipal bonds are exempt from federal taxation.
- The special feature in this paper is that *banks* do not receive the same tax treatment on all municipal bonds.
- If an issuer qualifies as a *qualified small issuer*, banks can hold its bonds with the favorable bank-qualified treatment.
- Before 2009, qualification required that the issuer's total tax-exempt issuance in the calendar year not exceed **\$10 million**; during 2009–2010 the limit was temporarily raised to **\$30 million**; after 2010 it reverted to \$10 million.

Interpretation. The policy does not directly subsidize municipalities. It subsidizes *banks that buy certain municipal bonds*. That is why the threshold can create segmentation between a bank-financed market and a non-bank market.

2.2 Tax treatment for banks

- When banks buy municipal bonds, coupon income is tax-exempt.
- But for *non-qualified* municipal bonds, banks lose the deduction for the interest expense incurred to finance those positions (the pro-rata disallowance).
- For *bank-qualified* bonds, only **20%** of the relevant interest expense is disallowed, so banks effectively preserve **80%** of that deduction.
- This discontinuity is specific to banks; households and non-bank institutional investors do not face this same qualification notch.

Interpretation. For banks, crossing the qualification threshold is like moving into a much worse after-tax pricing regime. That gives banks a strong reason to specialize in the bank-qualified segment.

2.3 Policy shock under ARRA

- In **February 2009**, Congress temporarily increased the bank-qualified limit from **\$10 million** to **\$30 million**.
- The stated goal was to expand small local governments' access to financing during the Great Recession.
- The reform lasted only for **2009 and 2010**, after which the threshold reverted.

Interpretation. The reform creates a sharp, temporary, and federally imposed change in access to bank-friendly municipal financing exactly when many local governments were fiscally stressed.

2.4 Data sources

- **Municipal bond issuance:** Ipreo MuniIC, with bond- and issue-level information since 2000.
- **Employment:** BLS QCEW census employment data.
- **Population:** Census.
- **House prices:** FHFA house price index.
- **Banks' balance sheets and municipal holdings:** Call Reports.
- **Deposits by geography:** FDIC Summary of Deposits.
- **Mortgage lending:** HMDA data from CFPB.
- **Small business lending:** CRA data from FFIEC.

Interpretation. The paper can follow the full chain from federal tax rules, to municipal issuance, to county employment, to banks' balance-sheet reallocations and local credit supply.

2.5 Main samples and time windows

- The **bunching** and pre-policy market-segmentation analysis uses municipal issuance from **2000–2008**.
- The main **county-level real-effects** analysis uses **2004–2010**.
- The main county panel contains **2,912 county-year observations**, corresponding to **416 counties**.
- The lending analysis begins later because regulatory coverage changes: CRA coverage is stable starting in **2005**, and the HMDA analysis spans **2006–2010**.
- For the real-effects design, the paper focuses on **single-county issuers** and excludes multi-county authorities and districts so that local spending is mapped to the right county.
- The preferred local-employment sample focuses on **urban counties** with population of at least **25,000**.

Interpretation. The paper deliberately narrows the sample when moving from municipal finance to local employment so that geographic spillovers and measurement error are less severe.

2.6 How treated and control municipalities are defined

- **Bunching issuers** are bank-qualified issuers located in the pre-2009 bunching region around the \$10 million cutoff. These are the issuers for whom the qualification threshold appears to be binding.
- **Control issuers** are smaller bank-qualified issuers that lie below the bunching region and therefore appear unconstrained by the \$10 million cap.
- The paper aggregates these issuers to the county level and defines

$$Intensity_i = \text{fraction of bunching municipalities in county } i. \quad (1)$$

- A county is *treated* if it has at least one bunching issuer, so $Intensity_i > 0$.

Interpretation. The treatment is not “large municipality” or “hard-hit county.” It is exposure to issuers whose borrowing behavior visibly distorted around the bank-qualification notch before the law changed.

2.7 Main county-level outcome variables

The county-level financing outcome is

$$Issuance_{i,t} = \text{bank-qualified debt raised in county } i \text{ at time } t, \quad (2)$$

where issuance sums debt raised by both bunching issuers and smaller bank-qualified controls.

Employment outcomes include:

$$\log(\text{Employment}_{i,t}), \quad \log(\text{PrivateEmployment}_{i,t}), \quad \log(\text{GovtEmployment}_{i,t}), \quad (3)$$

plus sectoral decompositions into goods and services and analogous wage outcomes.

Interpretation. The first stage asks whether bank-friendly municipal financing increased where constraints were likely to bind. The second stage asks what an extra dollar of that financing did for local labor markets.

2.8 Bank exposure measure for the lending analysis

The paper defines a Bartik-like measure of a bank's exposure to bank-qualified issuance:

$$\text{BankIntensity}_{i,t} = \frac{\sum_j \text{DepositShare}_{i,j} \times \text{BQIssuance}_{j,t}}{\text{Assets}_i}. \quad (4)$$

- $\text{DepositShare}_{i,j}$ is the share of bank i 's deposits in county j , fixed at the year before the policy change.
- $\text{BQIssuance}_{j,t}$ is the amount of bank-qualified debt issued in county j at time t .
- Assets are also fixed at the pre-policy level in the exposure measure.

Interpretation. Banks with larger deposit footprints in counties that issue more bank-qualified debt after the reform are predicted to experience a larger municipal-bond supply shock.

2.9 Descriptive facts

- In the main county sample, the average value of *Intensity* is **0.136**, with median **0.115**. That means roughly **13.6%** of issuers in a county are, on average, previously bunched issuers.
- Aggregate bank-qualified issuance by bunching and control issuers averages **\$23.48 million** per county-year, with median **\$14.22 million**.
- Average total county employment is about **137,298**; private employment averages **116,863**; local-government employment averages **13,157**.
- Service-sector employment averages **94,653**, more than four times goods-sector employment of **22,263**.
- Bunching issuers are much larger than non-bunching issuers in population and budget levels, but their *per capita* revenues, expenditures, and debt outstanding look broadly similar.

Interpretation. Treated counties are not simply counties with huge public sectors. The main treated-versus-control difference is exposure to previously constrained issuers, not obviously different local fiscal structure on a per capita basis.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: a tax-induced notch in municipal financing

In the paper’s stylized model, municipalities choose debt issuance while banks demand municipal bonds subject to a tax discontinuity at the qualification threshold.

A municipality’s bond supply can be written as

$$B_i^S = \zeta_i(1 - r)^\alpha, \quad (5)$$

where ζ_i captures the municipality’s financing needs and r is the equilibrium interest rate on its debt.

Bank demand is written as

$$B^B = [1 - (t + \Delta t \mathbf{1}\{B > B^*\})]r, \quad (6)$$

where B^* is the bank-qualification limit, t is the tax wedge below the threshold, and Δt is the extra tax penalty above the threshold.

The municipality’s budget set therefore has a notch:

$$G = \begin{cases} \Pi + B(1 - r(t_0)), & \text{if } B \leq B^*, \\ \Pi + B(1 - r(t_1)), & \text{if } B > B^*, \end{cases} \quad (7)$$

with $t_1 > t_0$.

Interpretation. Once issuance crosses the qualification limit, after-tax bank demand becomes less attractive, equilibrium yields rise, and the municipality’s effective budget set jumps downward. That creates a standard notch problem and generates bunching just below B^* .

3.2 Bunching estimation around the \$10 million cutoff

The paper estimates a counterfactual issuance density around the notch using binned issuer counts:

$$n_j = \sum_{i=0}^p \beta_i b_j^i + \sum_{k=L}^U \gamma_k \mathbf{1}\{b_k = b_j\} + \sum_{r \in R} \eta \mathbf{1}\{r \in R\} + e_j. \quad (8)$$

Here:

- b_j is the distance of issuance bin j from the qualification cutoff, measured in logs;
- the polynomial term fits the smooth counterfactual density;
- the excluded region $[L, U]$ isolates the distorted area around the cutoff;
- the round-number controls R absorb ordinary bunching at multiples of \$5 million.

The key behavioral object is the reduction in issuance by the marginal buncher:

$$\Delta B^* = \frac{\hat{D}}{\hat{h}_0(B^*)}, \quad (9)$$

where $\hat{D}$ is estimated excess mass just below the cutoff and $\hat{h}_0(B^*)$ is the estimated counterfactual density at the threshold.

Implementation details.

- The preferred specification uses a **13th-degree polynomial**.
- Bin width is **5%**, corresponding to **\$500k** steps.
- Standard errors are computed with **10,000 bootstrap iterations**.
- The excluded region is chosen so that missing mass and excess mass line up as closely as possible, which minimizes extensive-margin responses.

Interpretation. The bunching design answers two questions at once: *how many* issuers distorted their behavior and *by how much* the marginal constrained issuer cut issuance to keep bank-qualified status.

3.3 County-level first stage and second stage

The first stage is

$$Issuance_{i,t} = b_1 Intensity_i \times Post_t + b_2 X_{i,t} + a_i + a_{size,t} + e_{i,t}, \quad (10)$$

where $Post_t$ equals one in **2009–2010**, a_i are county fixed effects, and $a_{size,t}$ are population-decile-by-year fixed effects.

The second stage is

$$Y_{i,t} = \beta_1 \widehat{Issuance}_{i,t} + \beta_2 X_{i,t} + a_i + a_{size,t} + \xi_{i,t}, \quad (11)$$

where $Y_{i,t}$ is log employment or log wages.

Controls $X_{i,t}$ include county population and the house price index.

Interpretation. The instrument $Intensity_i \times Post_t$ compares counties with more versus fewer previously constrained issuers before and after the threshold expansion. The second stage then converts that induced extra issuance into local employment or wage effects.

3.4 Event-study / dynamic specification

To study timing and pre-trends, the paper estimates calendar-time interactions of treatment intensity with year dummies:

$$Y_{i,t} = \sum_{\tau \neq 2008} \delta_\tau (Intensity_i \times \mathbf{1}\{t = \tau\}) + \Gamma X_{i,t} + a_i + a_{size,t} + u_{i,t}, \quad (12)$$

with **2008** as the omitted base year in the main event-study figures.

Interpretation. This checks whether treated and control counties were already diverging before the reform and whether the response shows up only after the law changes.

3.5 Bank municipal-bond holdings regression

To show that banks exposed to newly available qualified issuance actually buy more municipal bonds, the paper estimates

$$\log(MuniBonds_{i,t}) = \beta BankIntensity_{i,t} + \gamma X_{i,t-1} + a_i + e_{i,t}. \quad (13)$$

The controls include lagged net income over assets, cost of deposits, equity over assets, and cash over assets, all measured at the bank level.

Interpretation. This is the bridge from the policy shock to banks' balance sheets. If the mechanism is right, banks that are more exposed through local deposit footprints should end up holding more municipal bonds after the reform.

3.6 Private-lending regression

To study crowding out, the paper estimates

$$\log(Loans_{i,j,t}) = \beta BankIntensity_{i,t} + \gamma X_{i,t-1} + a_i + \eta_{j,t} + e_{i,j,t}, \quad (14)$$

where $Loans_{i,j,t}$ is either CRA small-business lending or HMDA mortgage lending by bank i in county j at time t , and $\eta_{j,t}$ are county-by-year fixed effects.

Interpretation. County-by-year fixed effects absorb local credit demand. The identifying variation comes from comparing more-exposed and less-exposed banks lending in the *same county at the same time*.

3.7 How the job multiplier is computed

The employment regressions are in logs, so the paper converts the estimated elasticity into jobs using the average employment-to-debt ratio in the sample. In the preferred total-employment specification,

$$0.193 \times 112.42 \approx 21.7, \quad (15)$$

which the paper rounds to about **22 jobs per \$1 million** of spending.

The aggregate effect is then approximated by

$$22 \times 3,160 \approx 69,520, \quad (16)$$

where 3,160 is the extra bank-qualified debt in millions of dollars.

Interpretation. The 2SLS coefficient itself is an elasticity. The jobs-per-million figure is a trans-

parent translation of that elasticity into an economically interpretable multiplier.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the qualification threshold is a useful source of variation

- The threshold is embedded in federal tax law, not chosen by municipalities.
- Banks respond strongly to it: they mostly hold qualified rather than non-qualified bonds.
- Municipal spreads jump by about **10 basis points** above the cutoff, relative to an average spread of **132.7 basis points**, implying materially higher borrowing costs outside the bank-qualified segment.
- The bunching at \$10 million is strong before 2009 and drops sharply when the threshold is moved to \$30 million in 2009–2010.

Interpretation. The notch is not a symbolic legal detail. It affects investor clientele, prices, and issuer behavior in exactly the way the mechanism predicts.

4.2 Why treated and control counties are plausible counterfactuals

- Both treated and control issuers are in the *bank-qualified* segment, so they rely on the same investor type.
- The difference is whether the \$10 million limit appears to bind: bunchers are near the notch; controls are safely below it.
- Bunching issuers are larger in levels, but their per capita budgets look broadly similar to those of non-bunching qualified issuers.
- Pre-treatment county trends in employment, wages, and house prices are similar after adding population-decile-by-year fixed effects.

Interpretation. The design compares constrained and unconstrained issuers within the same financing regime, rather than comparing bank-financed municipalities to municipalities financed in an entirely different market segment.

4.3 Why the results are not just the housing bust or broader recession severity

- The paper controls for the county house price index and population.
- It uses county fixed effects and population-decile-by-year fixed effects to absorb persistent county differences and common time-varying shocks by county size.

- House-price event studies show no strong differential house-price dynamics between treated and control counties.
- The employment results are robust when excluding the states most affected by foreclosures in the sample: **Arizona, California, and Utah**.

Interpretation. The author cannot include state-by-year fixed effects because many states have too few treated and control counties, but the paper still works hard to show that the estimates are not simply picking up differential exposure to the housing crash.

4.4 Why the no-effect for larger newly eligible issuers matters

- Issuers in the \$10–\$30 million range became newly eligible after 2009.
- If the policy simply caused all newly eligible issuers to borrow more, one should see a strong response there as well.
- The paper finds essentially no increase in borrowing for those larger issuers.

Interpretation. This is an important scope condition. The reform mattered where the bank-financing constraint was binding, not where municipalities already had easy access to the traditional bond market.

4.5 Why the employment IV should be read as a local multiplier

- The 2SLS design identifies the employment effect of the *incremental debt issuance induced by the threshold change*.
- The estimate is therefore local to counties with more previously constrained issuers.
- The paper aggregates municipalities to the county level because town-level spending can spill across municipal borders; a municipality is too small an economic unit for a clean local-labor-market multiplier.

Interpretation. The estimate is not “the” national fiscal multiplier. It is the local employment effect of a very specific recession-era financing intervention for relatively small governments.

4.6 Why the bank crowding-out design is credible

- The bank exposure measure uses *pre-policy* deposit shares, so it does not mechanically reflect contemporaneous bank branch reallocation.
- County-by-year fixed effects absorb local loan demand in the lending regressions.
- The comparison is therefore between different banks lending in the same county-year, not between different counties.
- The paper also reports shift-share diagnostics and notes that influential counties still involve many distinct exposed banks, reducing concern that one or two institutions drive the results.

Interpretation. The crowding-out evidence is not based on raw cross-county correlations. It is based on within-county, across-bank differences in exposure to the municipal-bond supply shock.

4.7 What the paper cannot fully disentangle

- The paper cannot cleanly separate whether the extra debt financed short-run budget relief versus long-term capital projects, though the use-of-proceeds data suggest a mix of general purposes, education infrastructure, and water/sewer projects.
- Because the policy change was part of ARRA, other federal transfers may interact with local borrowing decisions.
- The paper argues that major transportation-related ARRA disbursements were relatively slow, which reduces but does not eliminate this concern.
- Mortgage crowding out is statistically meaningful, but the evidence on broader private-credit displacement is more moderate than in some sovereign-debt settings.

Interpretation. The paper is strongest on the existence of a financing channel and on the sign and magnitude of local employment effects. It is more cautious on the exact composition of spending and on the full general-equilibrium cost of crowding out.

5 Tables and figures

5.1 Fig. 1 and Fig. 3: market segmentation and borrowing costs

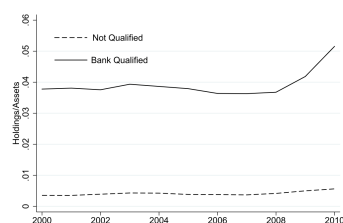


Fig. 1. Banker Holdings of Municipal Bonds.
This figure plots average banker holdings of bank-qualified (solid line) and non-qualified (dashed line) municipal bonds, as of December of the calendar year. Holdings are expressed as fraction of total assets. Data comes from Call Reports.

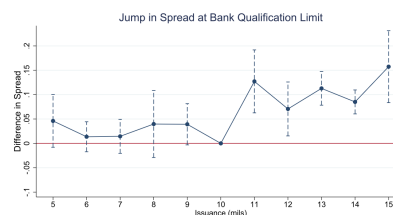


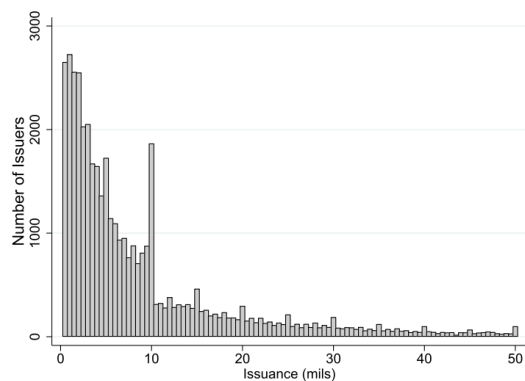
Fig. 3. Municipal Bond Spreads.
This figure shows the jump in spreads around the bank qualification cut-off. The sample includes general obligation bonds issued before 2008. Spreads are defined as tax-adjusted municipal bond yields over maturity-matched treasuries. On the x-axis is the size of the debt issuance. The coefficients are expressed as difference from the average spread of \$10M issuances. The dashed lines represent 90% confidence intervals.

Read.

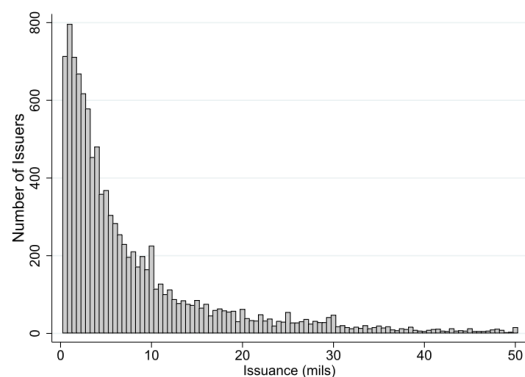
- Banks hold mostly bank-qualified municipal bonds: roughly **4–5% of total assets** versus just above **0.5%** for non-qualified bonds.
- For general-obligation bonds before 2008, spreads jump by about **10 basis points** once issuance moves above the bank-qualification cutoff.
- Since the average spread is **132.7 basis points**, the jump is economically meaningful — about a **7.5%** increase.

Economic message. The municipal bond market is segmented by investor type. Banks dominate the qualified segment, and municipalities pay noticeably more when they lose access to that segment.

5.2 Figs. 2, 4, and 5 plus Table 1: bunching around \$10 million



(a) Panel A: Issuance Distribution, 2000-2008



(b) Panel B: Issuance Distribution, 2009-2010

Fig. 2. Panel A plots the distribution of issuers for years 2000–2008. Panel B plots the distribution of issuers for years 2009–2010. On the x -axis is the size of the debt issuance per year, broken down in \$500k increments, while on the y -axis is the number of municipal issuers in each size bin.

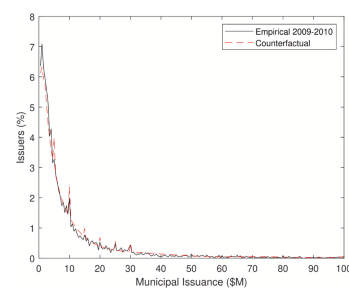


Fig. 4. Counterfactual Validation.
This figure plots the counterfactual distribution obtained using the bunching estimation described in Section 4 (in red) along with the observed issuance distribution for the period 2009–2010 (in black). The x -axis represents issuance. The y -axis is the fraction of issuers in each \$500k issuance size bin. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Read.

- Before 2009 there is a sharp spike in issuers just below **\$10 million** and missing mass just above the threshold.
- When the threshold rises to **\$30 million** in 2009–2010, the excess mass around \$10 million drops sharply.
- The preferred bunching estimate gives a behavioral response of **2.93%**.
- The preferred intensive-margin estimate implies about **41%** of issuers in the relevant missing region resized issuance to stay qualified.
- Extensive-margin effects are small and statistically insignificant.

Economic message. The policy does not merely affect who buys municipal debt. It changes how

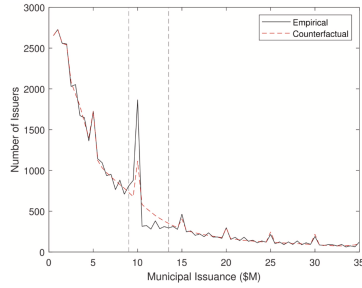


Fig. 5. Bunching Estimation.
This figure plots the observed distribution of issuers (solid black line) alongside the estimated counterfactual (dashed red line) for the years 2000–2008, around the bank qualification cut-off. The estimation follows the methodology described in Section 4. Each point represents the count of issuers in each 5% issuance size bin, while the vertical dashed line marks the region of exclusion.

Table 1
Bunching at the Bank Qualification Limit
This table reports the results of the bunching estimation from Section 4. Reported in the table are: the behavioral response (ΔB^*), the intensive margin effect ($\Delta B/N^*$), the extensive margin effect ($(\hat{M} - \hat{B})/N^*$), the lower and upper limits of the affected issuance region, and the choice of polynomial. Bootstrapped standard errors and 95% confidence intervals are in parentheses.

	(1)	(2)	(3)	(4)
Behavioral Response (ΔB^*)	0.0293***	0.0294***	0.0221***	0.0324***
<i>st. err.</i>	(0.0038)	(0.0043)	(0.0061)	(0.0039)
95% C.I.	(0.0221, 0.0375)	(0.0215, 0.0389)	(0.0117, 0.0365)	(0.0253, 0.0409)
Intensive Margin ($\Delta B/N^*$)	0.4137***	0.3447***	0.2937***	0.2963***
<i>st. err.</i>	(0.0634)	(0.0646)	(0.1333)	(0.0392)
95% C.I.	(0.3017, 0.5564)	(0.2362, 0.4983)	(0.106, 0.5509)	(0.2262, 0.3828)
Extensive Margin ($(\hat{M} - \hat{B})/N^*$)	-0.1758	-0.1487	-0.1444	0.0169
<i>st. err.</i>	(0.1195)	(0.1425)	(0.3885)	(0.0832)
95% C.I.	(-0.4418, 0.0288)	(-0.408, 0.0857)	(-1.0657, 0.2398)	(-0.1646, 0.1626)
Exclusion Limits (\$M)	(8.5, 13.5)	(8.5, 14.5)	(8, 17.5)	(9, 13.5)
Polynomial	13	13	13	13

much municipalities choose to issue in a year, and the main distortion is under-issuance rather than complete exclusion from capital markets.

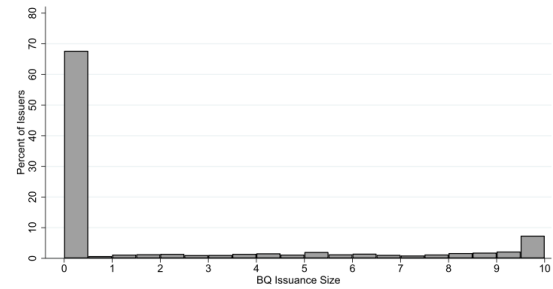
5.3 Table 2 and Fig. 6: borrowing dynamics

Table 2

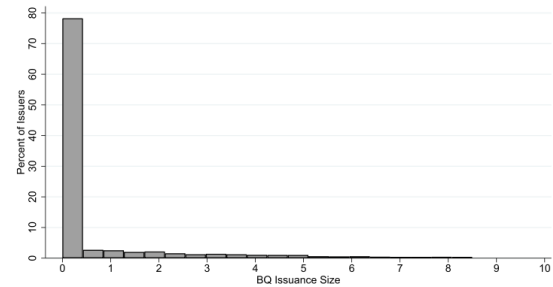
Dynamics of Issuance

This table presents the issuance patterns for both bunching and non-bunching (control) issuers. The variable *Number of Issuances* denotes the average number of times an issuer raised bank qualified debt during the 2000–2008 period. *Years between Issuances* represents the average time interval between consecutive debt issuances for issuers that raised debt more than once within this time period. Standard deviations are provided in parentheses.

	(1) Bunching issuers	(2) Controls	(3) Diff.
<i>Number of Issuances</i>	2.798 (1.802)	1.767 (1.256)	1.032*** std. err. = 0.009
<i>Years between Issuances</i>	1.789 (1.232)	2.059 (1.397)	-0.270*** std. err. = 0.023



(a) Panel A: Issuance Distribution at $t + 1$ - Bunching



(b) Panel B: Issuance Distribution at $t + 1$ - Controls

Fig. 6. This figure shows the distribution of issuance for bunching (Panel A) and non-bunching (Panel B) issuers, conditional on having issued the year before.

Read.

- Bunching issuers borrow more frequently: on average **2.798** times during 2000–2008 versus **1.767** times for controls.
- The average years between issuances are **1.789** for bunchers and **2.059** for non-bunchers.

- But dynamic smoothing is limited: conditional on issuing in the prior year, less than **10%** of bunchers bunch again next year, and about **67.5%** do not issue at all in the next year.

Economic message. Some municipalities may spread borrowing across years, but most bunching is not a trivial timing artifact. The threshold still imposes real financing costs through delayed investment or smaller issues.

5.4 Table 3, Fig. 7, and Table 4: treated versus control counties

Table 3
Pre-treatment characteristics
Panel A in this table shows local governments' budgets for a sample of hand-matched municipalities. Budget characteristics are as of the 2007 Census. Bunching and non-bunching qualified issuers are defined in Section 5.1. Panel B reports pre-trends for treated (*Intensity* > 0) and control (*Intensity* = 0) counties. Column (5) reports differences in estimated coefficients after controlling for population decline by year fixed effects. Robust standard errors clustered by county are reported in parentheses. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

	(1) Bunching	(2) Non-Bunching	(3) Difference	(4) St. Err.	(5) Diff. (adj.)	(6) St. Err.	(7) Obs.
Population	24,817	7,975	16,842.226***	(1,187.758)			3,029
Revenues (\$ ths)	43,902	12,962	30,939.648***	(2,626.351)			3,029
Intergov't Rev. (\$ths)	7,312	1,946	5,366.176***	(638.158)			3,029
Taxes (\$ths)	16,043	4,538	11,505.068***	(869.303)			3,029
Property Taxes (\$ths)	9,161	2,624	6,536.932***	(666.735)			3,029
Expenditures (\$ths)	45,596	13,483	31,912.863***	(2,565.879)			3,029
Debt Outstanding (\$ths)	43,856	13,438	30,418.511***	(3,208.893)			3,029
Revenues (per cap.)	1.965	1.894	0.161	(0.100)			3,029
Intergov't Rev. (per cap.)	0.289	0.292	-0.002	(0.024)			3,029
Taxes (per cap.)	0.722	0.592	0.130**	(0.060)			3,029
Property Taxes (per cap.)	0.428	0.372	0.055	(0.051)			3,029
Expenditures (per cap.)	2.053	1.961	0.092	(0.107)			3,029
Debt Outstanding (per cap.)	2.236	1.941	0.295	(0.184)			3,029

	(1) Treated	(2) Control	(3) Diff.	(4) St. Err.	(5) Diff. (adj.)	(6) St. Err.	(7) Obs.
<i>Δ</i> Employment	0.007	0.005	0.002	(0.002)	0.003	(0.002)	1,664
<i>Δ</i> Private Emp	0.006	0.005	0.001	(0.002)	0.002	(0.002)	1,660
<i>Δ</i> Gov Emp	0.011	0.009	0.003	(0.002)	0.003	(0.003)	1,475
<i>Δ</i> Goods Emp	-0.014	-0.006	-0.008*	(0.005)	-0.004	(0.005)	1,660
<i>Δ</i> Service Emp	0.012	0.009	0.003	(0.002)	0.003*	(0.002)	1,664
<i>Δ</i> Wages	0.045	0.044	0.001	(0.003)	0.003	(0.004)	1,664
<i>Δ</i> Private Wages	0.045	0.046	-0.001	(0.003)	0.001	(0.004)	1,660
<i>Δ</i> Gov Wages	0.046	0.045	0.001	(0.004)	0.001	(0.004)	1,516
<i>Δ</i> Goods Wages	0.032	0.040	-0.008	(0.006)	-0.006	(0.007)	1,660
<i>Δ</i> Service Wages	0.051	0.049	0.002	(0.003)	0.005	(0.004)	1,664
<i>Δ</i> HPI	0.020	0.027	-0.007***	(0.002)	-0.002	(0.003)	1,664
<i>Δ</i> Issuance	0.003	0.015	-0.012	(0.036)	-0.013	(0.041)	1,664

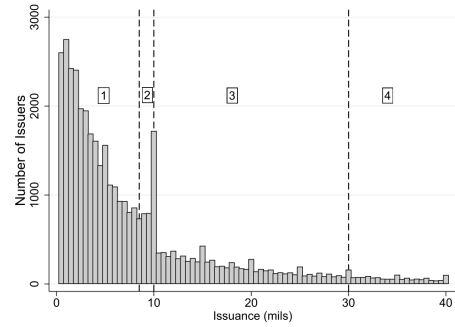


Fig. 7. Issuers Classification. This figure illustrates the classification of issuers across regions of the issuance distribution as described in Section 5.1.

Read.

- Bunching issuers are larger in raw population and budget levels, but per capita revenues, expenditures, and debt are similar to those of non-bunchers.
- Pre-treatment employment, wage, and house-price trends are broadly similar across treated and control counties once size-by-year fixed effects are included.
- In the county sample, mean *Intensity* is **0.136**; mean issuance is **\$23.48 million**; service employment is far larger than goods employment.

Table 4

Summary Statistics

This table presents summary statistics for the key variables used in the analysis. *Intensity* refers to the fraction of bunching municipalities within a county affected by the policy change, as detailed in Section 5.1. *Issuance* represents the total amount of bank-qualified debt issued by both bunching and control issuers in a year. Data on issuance, employment and wages span the years 2004–2010.

	(1) Obs.	(2) Mean	(3) St.Dev.	(4) Median
<i>Intensity</i>	2,912	0.136	0.121	0.115
<i>Issuance (\$ Mils.)</i>	2,912	23.48	35.10	14.22
<i>Employment - Total</i>	2,912	137,298	305,732	46,486
<i>Employment - Private</i>	2,912	116,863	265,280	38,336
<i>Employment - Goods</i>	2,912	22,263	45,759	8,161
<i>Employment - Services</i>	2,912	94,653	221,001	28,686
<i>Employment - Local Gvt</i>	2,912	13,157	30,620	5,002
<i>Wages - Total (\$Bils.)</i>	2,912	6.180	15.46	1.585
<i>Wages - Private (\$Bils.)</i>	2,912	5.253	13.41	1.289
<i>Wages - Goods (\$Bils.)</i>	2,912	1.257	2.902	0.370
<i>Wages - Service (\$Bils.)</i>	2,912	3.998	10.70	0.839
<i>Wages - Local Gvt (\$Bils.)</i>	2,912	0.561	1.552	0.165

Economic message. The design looks most credible when read as “more exposed versus less exposed to the threshold” rather than “fundamentally different counties.” The pre-period evidence supports that interpretation.

5.5 Table 5 and Fig. 8: first-stage issuance response

Table 5

Real Effects of Bank Qualification - Issuance

This Table reports first stage results from the 2SLS specification in Eq. (6). *Intensity* is the fraction of bunching municipalities in a county. *Post* takes value of one during the regulatory change (2009–2010), and zero otherwise (2004–2008). *Issuance* is the bank-qualified debt raised in a county each year as specified in Section 5.1, and expressed in \$100Mils. Controls include HPI and population. Robust standard errors clustered by county are reported in parenthesis. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

Dep. Var.:	(1) Issuance	(2) Issuance	(3) Issuance
<i>Intensity</i> × <i>Post</i>	0.297*** (0.080)	0.293*** (0.080)	0.224*** (0.072)
<i>Post</i>	0.012 (0.010)	0.015 (0.010)	
Controls	No	Yes	Yes
County FE	Yes	Yes	Yes
Size Decile × Year FE	No	No	Yes
Observations	2,912	2,912	2,912
Adjusted R-squared	0.828	0.829	0.835

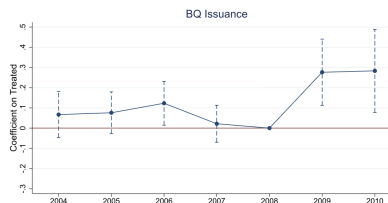


Fig. 8. Effects on Issuance.

This figure illustrates the effects on municipal bond issuance using a calendar-time coefficient plot. The plot shows the regression coefficients of the logarithm of *Issuance* on the interaction between treated *Intensity* and year dummies, covering the period from 2004 to 2010. The year 2008 serves as the base year. The dashed lines correspond to 90% confidence intervals.

Read.

- In the preferred specification, the coefficient on *Intensity* × *Post* is **0.224** when issuance is measured in **\$100 million** units.
- Multiplying by the standard deviation of intensity, **0.121**, implies an increase in issuance of about **\$2.7 million**.
- The first-stage *F*-statistic stays around **9.6** in the most stringent specification.
- The event-study figure shows the increase in issuance appearing only after the 2009 reform, not before.

Economic message. Expanding the threshold relaxed a genuinely binding financing constraint for previously bunched issuers.

5.6 Tables 6–8 and Figs. 9–10: employment effects and robustness

Table 6

Real Effects of Bank Qualification - Employment

This Table reports second stage results from the 2SLS specification in Eq. (7). The dependent variable is the log of employment. *Intensity* is the fraction of bunching municipalities in a county. *Post* takes value of one during the regulatory change (2009–2010), and zero otherwise (2004–2008). *Issuance* is the bank-qualified debt (expressed in \$100Mils) raised in a county each year as specified in Section 5.1, and is instrumented using *Intensity* × *Post*. Controls include HPI and population. Robust standard errors clustered by county are reported in parenthesis. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

Dep. Var.:	(1) Log(Employment)	(2)	(3)	(4) Log(Private Employment)	(5)	(6)	(7) Log(Govt Employment)	(8)	(9)
<i>Issuance</i>	0.188** (0.087)	0.158* (0.082)	0.193* (0.116)	0.231** (0.099)	0.199** (0.094)	0.219* (0.131)	0.048 (0.091)	0.019 (0.090)	0.058 (0.107)
<i>Post</i>	-0.001*** (0.004)	-0.040*** (0.004)		-0.002*** (0.005)	-0.003*** (0.005)		0.015** (0.008)	0.016** (0.008)	
Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size Decile × Year FE	No	No	Yes	No	No	Yes	No	No	Yes
F-stat 1st stage	13.78	13.35	9.621	13.66	13.23	9.542	15.83	15.35	12.88
Observations	2,912	2,912	2,912	2,905	2,905	2,905	2,622	2,622	2,622
Counties	416	416	416	415	415	415	382	382	382

Table 7

Private Sector Employment- Goods vs. Services

This Table reports second stage results from the 2SLS specification in Eq. (7). The dependent variable is the log of private employment, broken down in goods and service sectors. *Issuance* is the instrumented bank-qualified debt (expressed in \$100Mils) raised in a county each year as specified in Section 5.1. Controls include HPI and population. Robust standard errors clustered by county are reported in parenthesis. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

Dep. Var.:	(1) Log(Goods Emp)	(2) Log(Service Emp)
<i>Issuance</i>	-0.232 (0.189)	0.390** (0.170)
Controls	Yes	Yes
County FE	Yes	Yes
Size Decile × Year FE	Yes	Yes
F-stat 1st stage	9.542	9.621
Observations	2,905	2,912
Counties	415	416

Table 8

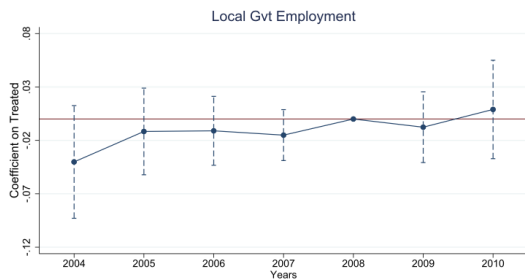
Excluding High Foreclosure States

This Table reports second stage results from the 2SLS specification in Eq. (7). This sample excludes the states most affected by the foreclosure crisis. *Issuance* is the instrumented bank-qualified debt (expressed in \$100Mils) raised in a county each year as specified in Section 5.1. Controls include HPI and population. Robust standard errors clustered by county are reported in parenthesis. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

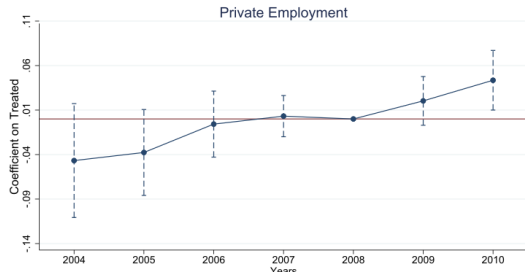
Dep. Var.:	(1) Log(Emp)	(2) Log(Priv. Emp)	(3) Log(Govt Emp)
<i>Issuance</i>	0.190* (0.115)	0.221* (0.130)	0.044 (0.105)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Size Decile × Year FE	Yes	Yes	Yes
F-stat 1st stage	10.11	10.03	13.39
Observations	2,821	2,814	2,533
Counties	403	402	369

Read.

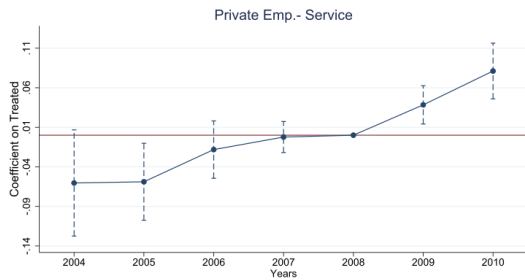
- The preferred second-stage coefficient is **0.193** for total employment and **0.219** for private employment.
- Using the paper's conversion, that is about **22 jobs** per \$1 million overall and just under **21 private-sector jobs** per \$1 million.
- The government-employment coefficient, **0.058**, is small and statistically insignificant.
- In the private-sector decomposition, the goods coefficient is **-0.232** (insignificant) and the services coefficient is **0.390****.
- Results remain similar after excluding the high-foreclosure states in the sample.
- The house-price event study does not show strong differential trends.



(a) Panel A: Government Employment



(b) Panel B: Private Employment



(c) Panel C: Private Employment - Services



Fig. 10. House Price Index.
This figure reports the regression coefficients of the logarithm of the house price index on the interaction between treated and year dummies, covering the period from 2004 to 2010, with 2004 as the base year.

Fig. 9. This figure illustrates the effects on employment using a calendar-time coefficient plot. The plot shows the regression coefficients of the logarithm of employment by sector on the interaction between treated *Intensity* and year dummies, covering the period from 2004 to 2010. The year 2008 serves as the base year. The dashed lines correspond to 90% confidence intervals.

Economic message. The real effect is not “more public payroll” in a mechanical sense. The evidence points to broader private-sector job creation in local services, consistent with municipal spending supporting local demand and local contractors.

5.7 Tables 9–10 and Fig. 11: bank holdings and crowding out

Table 9

Bank Holdings of Municipal Bonds

This Table reports results from the specification in Eq. (9). $\text{Log}(\text{Muni Bonds})$ is the log of municipal bond holdings, measured at the bank holding company level. BankIntensity is the standardized bank's exposure to municipal bond issuance as defined in Eq. (8). $\text{NetIncomeRatio}_{t-1}$, $\text{CostofDeposits}_{t-1}$, EquityRatio_{t-1} , CashRatio_{t-1} are measured at the bank holding company level and are respectively net income over total assets, interest expense on deposits over deposits, equity over total assets, and cash over total assets, each lagged by one year. Robust standard errors are clustered at the bank level and are reported in parenthesis. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

Dep. Var.:	(1)	(2)	(3)
	Log(Muni Bonds)	Log(Muni Bonds)	Log(Muni Bonds)
<i>Bank Intensity</i>	0.095* (0.053)	0.094* (0.050)	0.071** (0.036)
<i>NetIncomeRatio</i> _{t-1}		0.424* (0.219)	0.188 (0.197)
<i>CostofDeposits</i> _{t-1}		-0.061 (0.047)	-0.054 (0.043)
<i>EquityRatio</i> _{t-1}			-2.568*** (0.371)
<i>CashRatio</i> _{t-1}			1.074*** (0.413)
Bank FE	Yes	Yes	Yes
Observations	25,683	25,379	25,379
Adjusted R-squared	0.841	0.843	0.843

Table 10

Private Lending

This Table reports results from the reduced form specification (10). $\text{Log}(\text{CRA Loans})$ is the log of a bank's new CRA loans extended in a county. $\text{Log}(\text{HMDA Origin})$ is the log of a bank's new mortgages originated in a county. $\text{Log}(\text{HMDA Purchased})$ is the log of a bank's new mortgages purchased in a county. BankIntensity is the standardized bank's exposure to municipal bond issuance as defined in Eq. (8). $\text{NetIncomeRatio}_{t-1}$, $\text{CostofDeposits}_{t-1}$, EquityRatio_{t-1} , CashRatio_{t-1} are respectively net income over total assets, interest expense on deposits over deposits, equity over total assets, and cash over total assets, each lagged by one year. Variables are measured at the bank holding company level. Robust standard errors are double clustered by county and bank, and are reported in parenthesis. Significance follows: * 10 percent; ** 5 percent; *** 1 percent.

Dep. Var.:	(1)	(2)	(3)
	Log(CRA Loans)	Log(HMDA Origin)	Log(HMDA Purchased)
<i>Bank Intensity</i>	-0.187 (0.620)	-0.189* (0.108)	0.102 (0.162)
<i>NetIncomeRatio</i> _{t-1}	2.691 (2.098)	0.268 (1.900)	2.572 (2.499)
<i>CostofDeposits</i> _{t-1}	4.899*** (1.077)	6.844 (5.985)	-7.639 (9.010)
<i>EquityRatio</i> _{t-1}	0.032 (2.931)	2.086 (2.266)	-3.379 (3.287)
<i>CashRatio</i> _{t-1}	-1.816 (2.150)	-3.424 (2.724)	4.155 (3.333)
Bank FE	Yes	Yes	Yes
County × Year FE	Yes	Yes	Yes
Observations	281,469	428,215	428,215
Adjusted R-squared	0.315	0.396	0.660

Read.

- In the preferred bank-holdings regression, a one-standard-deviation increase in bank exposure (**1.09**) predicts about a **7.7%** increase in municipal bond holdings.
- The increase in municipal holdings is strongest for **small and medium-sized banks**; small banks nearly double holdings after 2009 in the figure.
- In the lending regressions, the mortgage-origination coefficient is **-0.189***. With a one-standard-deviation exposure change of **0.38**, this implies about a **7%** fall in mortgage origination.
- The CRA small-business lending coefficient is negative but very imprecise, and purchased mortgages show no clear effect.

Economic message. The policy worked by redirecting bank balance sheets toward municipal bonds. That helped municipalities, but it was not free: some mortgage lending appears to have been displaced.

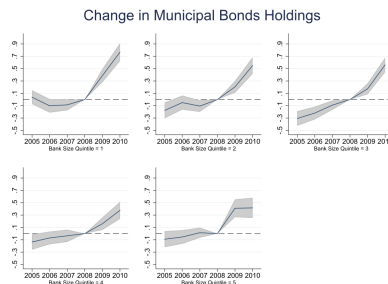


Fig. 11. Banks' Change in Muni Bond Holdings by Quintile.
This figure plots the regression coefficients of the logarithm of municipal bond holdings against year dummies for the period 2005–2010, with 2008 as the base year, broken down by banks' asset size quintiles. The shaded areas correspond to 90% confidence intervals.

5.8 Section 7 aggregate magnitudes: how large was the policy overall?

Read.

- The paper estimates that affected municipalities issued about **\$3.16 billion** more bank-qualified debt nationwide because of the reform.
- Applying the estimated multiplier of **22 jobs per \$1 million** gives roughly **70,000 jobs** created or saved.
- The implied cost per job-year is about **\$45,500**.
- At the county level, the paper's back-of-the-envelope calculation implies about **252 jobs** created per average county and **55 jobs** per median county.

Economic message. The policy is modest next to giant federal spending programs, but within the universe of small local governments it is economically meaningful and delivers multipliers in the same broad range as much larger interventions.

6 Short summary

- The bank-qualification rule created a distinct bank-financed municipal bond segment with lower borrowing costs.
- Municipalities responded by bunching just below the \$10 million qualification limit before 2009.
- Raising the threshold to \$30 million in 2009–2010 increased municipal borrowing in counties with more previously constrained issuers.
- The extra borrowing generated about **22 jobs per \$1 million**, mainly in the private service sector.
- Larger newly eligible issuers in the \$10–\$30 million range did not borrow more, consistent with the idea that they were not previously constrained.
- Banks exposed to the new municipal-bond supply increased muni holdings and reduced mortgage origination, providing moderate evidence of crowding out.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that investor-side tax policy can relax local governments' financing constraints during a recession.
- The bank-qualification threshold clearly shaped municipal issuance before 2009, and the temporary ARRA expansion loosened that constraint for affected small issuers.

- The resulting increase in municipal borrowing had measurable real effects on local employment.

7.2 Economic intuition

- A municipality that stays below the qualification cutoff can borrow in a bank-dominated segment with better after-tax pricing.
- When the cutoff rises, more of that issuer's financing can be placed with banks, reducing financing pressure and supporting spending.
- Because banks' balance sheets are finite, part of the policy works by redirecting credit capacity away from other assets, especially some mortgage lending.

7.3 Takeaway

- Credit-market design and tax rules for investors can matter as much as direct transfers when evaluating recession policy.
- The paper also shows why it is useful to study *market segmentation* rather than only average borrowing costs: the relevant margin here is who can access bank capital.
- More broadly, the note's bottom line is that tax incentives for banks to hold municipal debt can support small local economies, but policy evaluation should account for both the employment gains and the balance-sheet crowding out they generate.